CHAPTER 4 APPLICATIONS OF DIFFERENTIATION

4.1 Maxima and Minima

- **Definition**

Let D, the domain of f, contain the point c. We say that

(i) $f(c)$ is the **absolute maximum** value of f on D if $f(c) \geq f(x)$ for all x in D;

(ii) $f(c)$ is the **absolute minimum** value of f on D if $f(c) \leq f(x)$ for all x in D;

(iii) $f(c)$ is an absolute extreme value of f on D if it is either the absolute maximum value or the absolute minimum value;

C can be in the endpoint

- **Definition**

Let D, the domain of f, contain the point c. We say that

(i) $f(c)$ is a **local maximum** value of f if $f(c) \geq f(x)$ when x is near c;

(ii) $f(c)$ is a **local minimum** value of f if $f(c) \leq f(x)$ when x is near c;

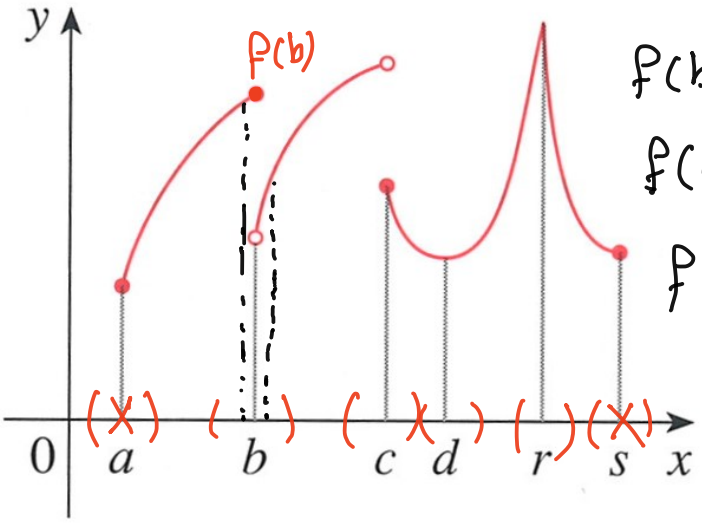
(iii) $f(c)$ is a local extreme value of f on D if it is either a local maximum value or a local minimum value.

C cannot be in the endpoint

Example 4.1. Find the absolute extreme values and the local extreme values of the function.

f(r) is absolute maximum value

f(a) is absolute minimum value



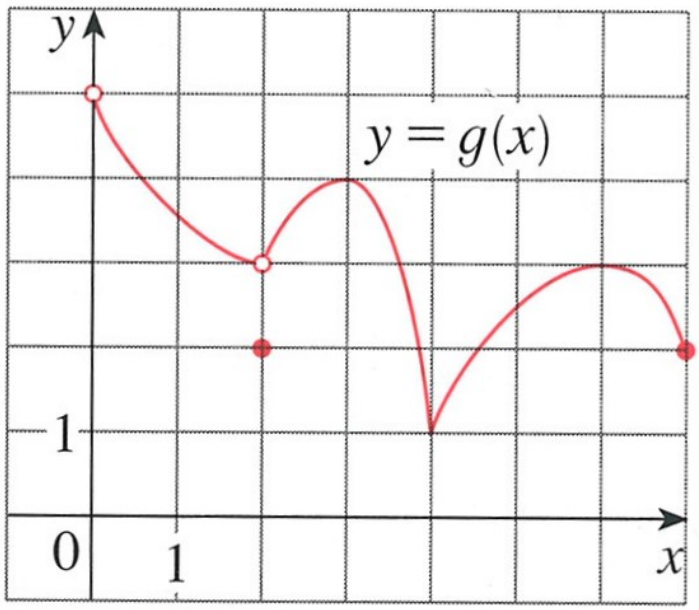
f(b) is local max. value

f(d) is local mini. value

f(r) is local max. value

D = [a , s]

Example 4.2. Find the absolute extreme values and the local extreme values of the function.



- **Extreme Value Theorem**

> If f is continuous on a closed interval $[a, b]$, then f attains both an absolute maximum value and an absolute minimum value there.

- If c is a point at which $f'(c) = 0$, we call c a **stationary point**.

Example 4.3. Find the stationary points of the function $f(x) = 2x^3 - 3x^2 + 2$.

$f'(x) = 6x^2 - 6x$

let $f'(x) = 0 \iff 6x^2 - 6x = 0$

$\iff 6x(x-1) = 0$

Either $x = 0$ or $x = 1$

Then, stationary points are: $(0, 2)$ & $(1, 1)$

- If c is an interior point of the domain of f such that $f'(c)$ does not exist, we call c a **singular point**.

$c \in (a, b)$ $\quad = \mp\infty$

Example 4.4. Find the singular points of the function $y = \sqrt[3]{x^2}$. $= x^{\frac{2}{3}}$

$y' = \frac{2}{3} x^{-\frac{1}{3}} = \frac{2}{3\sqrt[3]{x}}$

when $3\sqrt[3]{x} = 0 \iff \sqrt[3]{x} = 0 \iff x = 0$

Then, $y'(0)$ D.N.E

So, $(0, 0)$ is the singular point

- If f is a function defined on an interval, a **critical point** c of f is one of the following three types of points:
  (a) an end point of I; $[a, b]$
  (b) a stationary point of f; $f'(c) = 0$
  (c) a singular point of f. $f'(c)$ D.N.E, $c \in (a, b)$

- **Fermat's Theorem**

> Let f be defined on an interval I containing the point c. If $f(c)$ is a local extreme value, and $f'(c)$ exists, then $f'(c) = 0$. , c is stationary point

- **Critical Point Theorem**

absolute or local

> Let f be defined on an interval I containing the point c. If $f(c)$ is an extreme value, then c must be a critical point.

absolute

- In order to find the extreme values of a function f that is continuous on a closed interval $I = [a, b]$,
  Step 1 Find the critical points of f on I.
  Step 2 Evaluate f at each of these critical points. The largest value is the absolute maximum value; the smallest value is the minimum value.

absolute

Example 4.5. Find the absolute maximum and absolute minimum values of the function $f(x) = x^3 - 3x^2 + 1$ on $\left[-\frac{1}{2}, 4\right]$.

① End points are $x = -\frac{1}{2}$, $x = 4$

$f'(x) = 3x^2 - 6x$

② let $f'(x) = 0 \iff 3x^2 - 6x = 0 \iff 3x(x-2) = 0$

Either $x = 0 \in (-\frac{1}{2}, 4)$ or $x = 2 \in (-\frac{1}{2}, 4)$

③ $f'(x)$ Exist for $\forall x \in (-\frac{1}{2}, 4)$

∴ No singular points

Example 4.6. Find the absolute maximum and absolute minimum values of $F(x) = x^{\frac{2}{3}}$ on $[-1, 2]$.

$\frac{a}{b} = 0 \iff a = 0$

$F'(x) = \frac{2}{3} x^{-\frac{1}{3}} = \frac{2}{3\sqrt[3]{x}}$

① let $F'(x) = 0 \iff 2 \neq 0$

∴ There is no stationary point

② Singular points, when $3\sqrt[3]{x} = 0$

$\iff x = 0 \in (-1, 2)$

③ Endpoints :: $x = -1$, $x = 2$

⇓

Example (4.5) ∘∘

$f(-\frac{1}{2}) = (-\frac{1}{2})^3 - 3(-\frac{1}{2})^2 + 1 = \frac{1}{8}$

$f(4) = 4^3 - 3(4)^2 + 1 = 17$

$f(0) = 0 - 0 + 1 = 1$

$f(2) = 2^3 - 3(2)^2 + 1 = -3$

$\therefore f(4) = 17$ is the absolute max. value

& $f(2) = -3$ is the absolute mini. value

Example 4.6 ∘∘

$f(0) = 0$

$f(-1) = \sqrt[3]{1} = 1$

$f(2) = \sqrt[3]{4} = 1.587$

$\therefore f(2) = 1.587$ is absolute max. value

$f(0) = 0$ is absolute mini. value

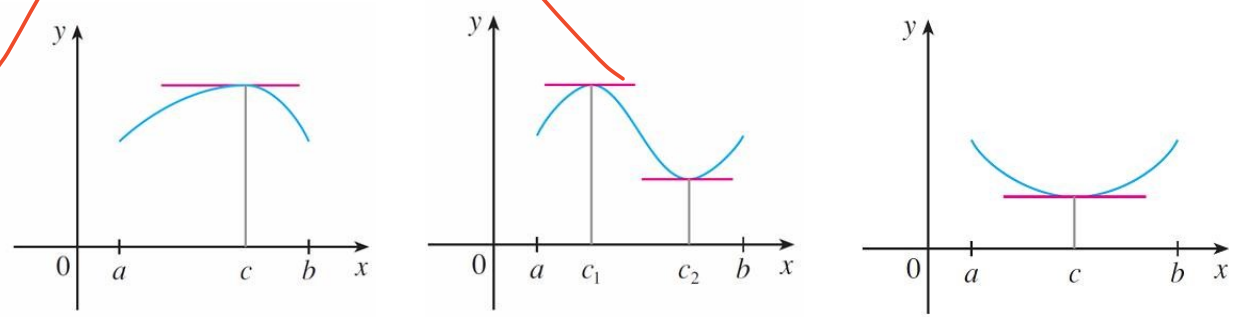
Example 4.7. Find the absolute maximum and absolute minimum values of $f(x) = x + 2\cos x$ on $[-\pi, 2\pi]$.

Exercises Page 283 5, 17, 27, 47, 53, 57.

4.2 The Mean Value Theorem for Derivatives

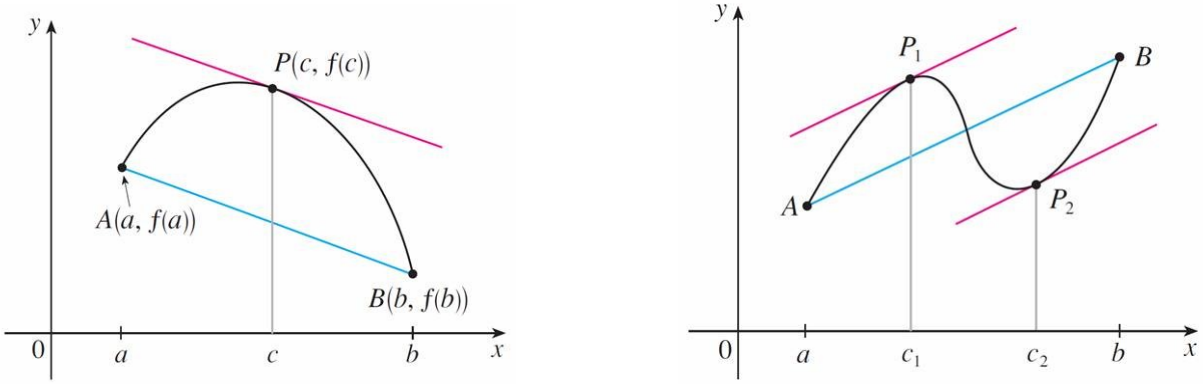
- **Rolle's Theorem**

> If f is continuous on a closed interval $[a, b]$ and differentiable on its interior (a, b), and $f(a) = f(b)$, then there is a c in (a, b) where
>
> $$f'(c) = 0$$



- **The Mean Value Theorem for Derivatives**

> If f is continuous on a closed interval $[a, b]$ and differentiable on its interior (a, b), then there is a c in (a, b) where
>
> $$f'(c) = \frac{f(b) - f(a)}{b - a}$$



Example 4.8. Find the number c guaranteed by the Mean Value Theorem for $f(x) = 2\sqrt{x}$ on $[1, 4]$.

Example 4.9. Let $f(x) = x^3 - x^2 - x + 1$ on $[-1, 2]$. Find all numbers c satisfying the conclusion of the Mean Value Theorem.

Example 4.10. Use the Intermediate Value Theorem and the Mean Value Theorem to prove that the equation $x^3 + x - 1 = 0$ has exactly one real root .

Example 4.11. Let $f(x) = x^{\frac{2}{3}}$ on $[-8, 27]$. Show that the conclusion of the Mean Value Theorem fails and figure out why.

Example 4.12. Suppose that an object has position $s(t) = t^2 - t - 2$. Find the average velocity over the interval $[3, 6]$ and find the time at which the instantaneous velocity equals the average velocity.

Example 4.13. If $f(0) = -3$ and $f'(x) \leq 5$ for all x, how large can $f(2)$ possibly be?

- **Theorem**

If $F'(x) = 0$ for all x in (a, b), then $F(x)$ is a constant C on (a, b).

If $F'(x) = G'(x)$ for all x in (a, b), then there is a constant C such that

$$F(x) = G(x) + C$$

for all x in (a, b).

Example 4.14. Prove that $\sin^{-1} x + \cos^{-1} x = \dfrac{\pi}{2}$.

Exercises Page 291 1, 3, 7, 9, 13, 17, 19, 21, 25, 29, 33.

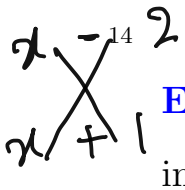
4.3 Monotonicity and Concavity

- **Monotonicity Theorem**

Let f be continuous on an interval I and differentiable at every interior point of I.

(i) If $f'(x) > 0$ for all x interior to I, then f is **increasing** on I.

(ii) If $f'(x) < 0$ for all x interior to I, then f is **decreasing** on I.



Example 4.15. If $f(x) = 2x^3 - 3x^2 - 12x + 7$, find the intervals on which f is increasing or decreasing.

$f'(x) = 6x^2 - 6x - 12$

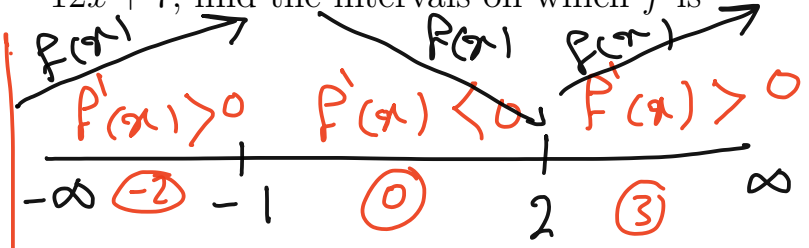
When $f'(x) = 0$

$6x^2 - 6x - 12 = 0$

$x^2 - x - 2 = 0$

$(x-2)(x+1) = 0$

Either $\boxed{x = 2}$ or $\boxed{x = -1}$



$f'(-2) = 6(-2)^2 - 6(-2) - 12$

∴ The function is increasing on

$(-\infty, -1)$, $(2, \infty)$

& the function is decreasing on

$(-1, 2)$

Example 4.16. On what intervals is $g(x) = \dfrac{x}{1+x^2}$ increasing? On what intervals is it decreasing?

increasing on $(-1, 1)$

decreasing on $(-\infty, -1)$, $(1, \infty)$

- **Local Extreme Values – First Derivative Test**

Let f be continuous on an open interval (a, b) that contains a critical point c.

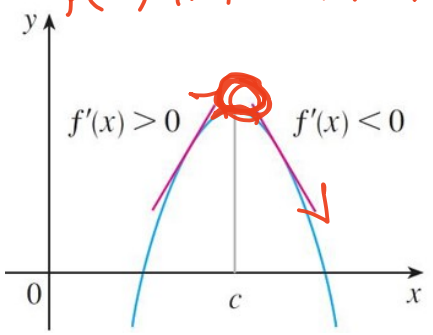
Stationary Singular

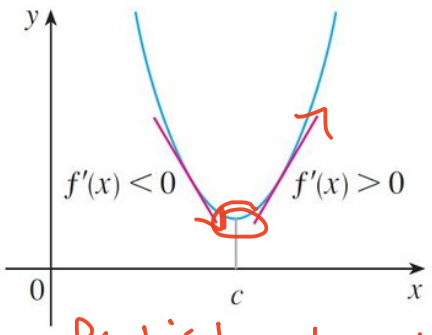
(i) If $f'(x) > 0$ for all x in (a, c) and $f'(x) < 0$ for all x in (c, b), then $f(c)$ is a **local maximum** value of f.

(ii) If $f'(x) < 0$ for all x in (a, c) and $f'(x) > 0$ for all x in (c, b), then $f(c)$ is a **local minimum** value of f.

(iii) If $f'(x)$ has the same sign on both sides of c, then $f(c)$ is not a local extreme value of f.

f(c) is local max.





f(c) is local mini.

Example 4.17. Find the local extreme values of $f(x) = x^3 - 3x^2 - 9x + 12$ on $(-\infty, \infty)$ using the First Derivative Test.

$f'(x) = 3x^2 - 6x - 9$

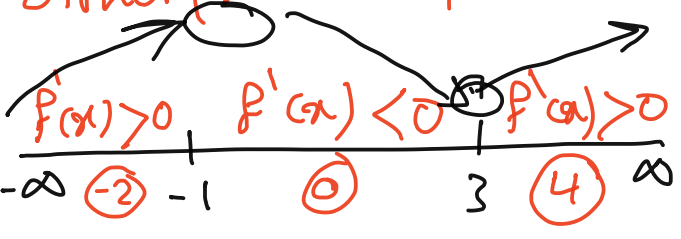
When $f'(x) = 0$

$3x^2 - 6x - 9 = 0$

$x^2 - 2x - 3 = 0$

$(x-3)(x+1) = 0$

Either $x = 3$ or $x = -1$



$f(-1) = (-1)^3 - 3(-1)^2 - 9(-1) + 12 = 17$ is local max. value

$f(3) = -15$ is local mini. value

HW

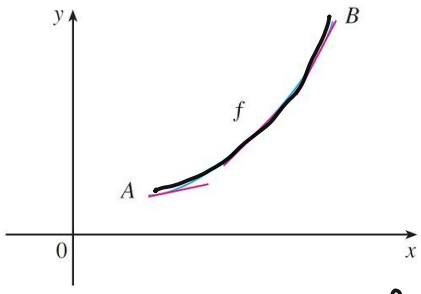
Example 4.18. Find the local extreme values of $f(x) = x + 2\sin x$ on $0 \le x \le 2\pi$ using the First Derivative Test.

$f\left(\frac{4\pi}{3}\right) \approx 2.46$ is local mini. value

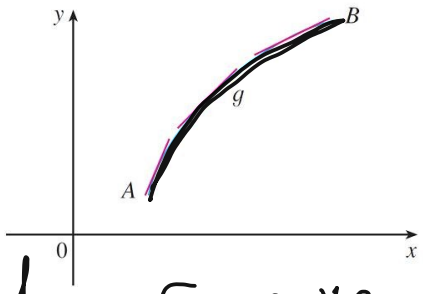
$f\left(\frac{2\pi}{3}\right) \approx 3.83$ is local max. value

- **Definition**

If the graph of f lies above all its tangent lines on an interval I, then we say that f is **concave upward** on I. If the graph of f lies below all its tangent lines on an interval I, then we say that f is **concave downward** on I.



Concave upward



Concave downward

Let f be differentiable on an open interval I. We say that f is **concave up** on I if f' is increasing on I, and we say that f is **concave down** on I if f' is decreasing on I.

when $f'' > 0$

when $f'' < 0$

- **Concavity Theorem**

Let f be twice differentiable on an open interval I.

(a) If $f''(x) > 0$ for all x in I, then f is **concave up** on I.

(b) If $f''(x) < 0$ for all x in I, then f is **concave down** on I.

Example 4.19. On what intervals is $f(x) = x^3 - 3x^2 - 9x + 12$ concave upward or concave downward?

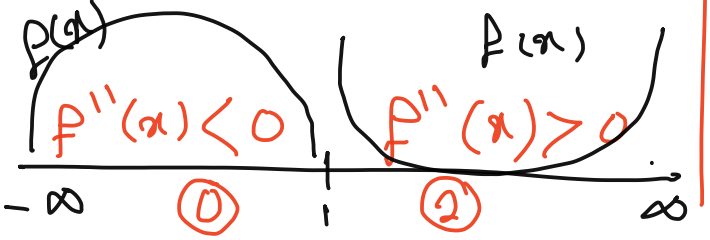
$f'(x) = 3x^2 - 6x - 9$

$f''(x) = 6x - 6$

When $f''(x) = 0$

$6x - 6 = 0$

$x - 1 = 0 \Leftrightarrow \boxed{x = 1}$



$f(x)$ is concave downward on $(-\infty, 1)$

$f(x)$ is concave upward on $(1, \infty)$

Example 4.20. Given the function $g(x) = \dfrac{x}{1+x^2}$. Find the intervals of concavity.

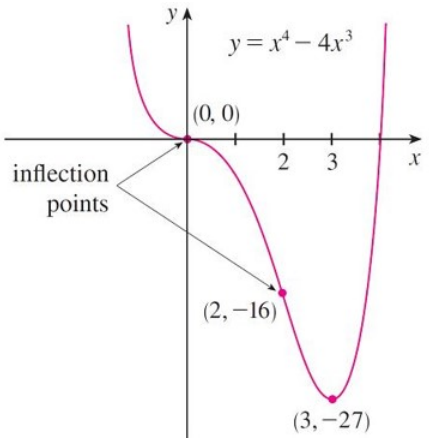
HW

Concave up on $(-\sqrt{3}, 0)$

Concave down on $(-\infty, -\sqrt{3})$, $(0, \sqrt{3})$

- **Inflection Points**

  We call $(c, f(c))$ an **inflection point** of the graph f if f is concave up on one side of c and concave down on the other side.



- If $(c, f(c))$ is an inflection point, then either $f''(c) = 0$ or $f''(c)$ does not exist.

Example 4.21. Find the inflection points of $F(x) = x^{\frac{1}{3}} + 2$.

$F'(x) = \frac{1}{3} x^{-\frac{2}{3}}$

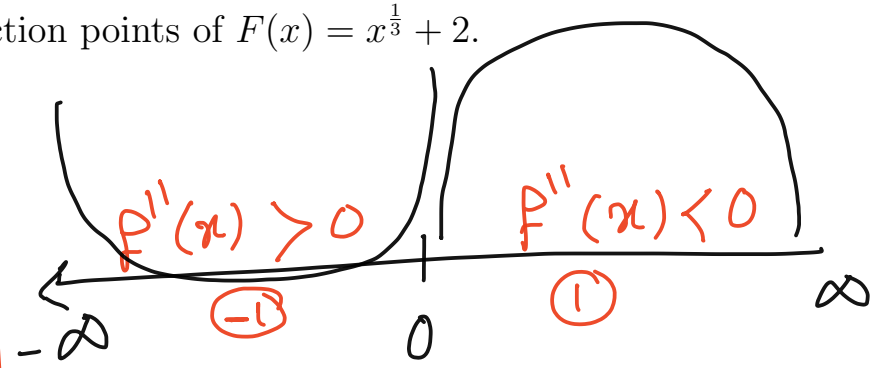
$F''(x) = \frac{-2}{9} x^{-\frac{5}{3}}$

$F''(x) = \frac{-2}{9\sqrt[3]{x^5}}$

When $F''(x) \neq 0$

When $F''(x)$ D.N.E

$9\sqrt[3]{x^5} = 0 \iff x = 0$



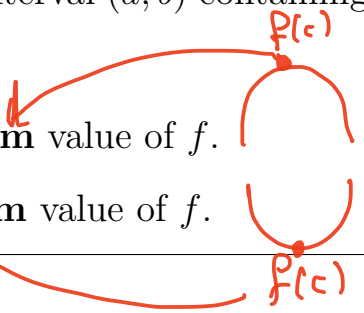
$\therefore (0, f(0)) = (0, 2)$ is the inflection point

- **Local Extreme Values – Second Derivative Test**

Let f' and f'' exist at every point in an open interval (a, b) containing c, and suppose $f'(c) = 0$.

(i) If $f''(c) < 0$, then $f(c)$ is a **local maximum** value of f.

(ii) If $f''(c) > 0$, then $f(c)$ is a **local minimum** value of f.



Example 4.22. Find (if any) the local maximum and minimum values of $f(x) = x^4 - 4x$ on $(-\infty, \infty)$ using the Second Derivative Test.

$f'(x) = 4x^3 - 4$

When $f'(x) = 0$

$4x^3 - 4 = 0 \iff (x^3 - 1) = 0$

$x^3 = 1 \iff \boxed{x = 1}$

$f''(x) = 12x^2$

$f''(1) = 12 > 0$ ∪

$\therefore$ $f(1) = -3$ is local mini. value

There is no local max.

HW

Example 4.23. Find (if any) the local maximum and minimum values of

$$G(p) = \frac{1}{p(1-p)}$$

on $(0, 1)$ using the Second Derivative Test.

$G(\frac{1}{2}) = 4$ is local min. value

Example 4.24. Given the function $f(x) = x^{\frac{2}{3}}(6-x)^{\frac{1}{3}}$.

(a) Find the intervals on which f is increasing or decreasing.

$$f'(x) = x^{\frac{2}{3}}\left(\frac{1}{3}\right)(6-x)^{-\frac{2}{3}}(-1) + (6-x)^{\frac{1}{3}}\left(\frac{2}{3}\right)(x)^{-\frac{1}{3}}$$

$$f'(x) = \frac{-x^{\frac{2}{3}} \times x^{\frac{1}{3}}}{3(6-x)^{\frac{2}{3}}x^{\frac{1}{3}}} + \frac{2(6-x)^{\frac{1}{3}}(6-x)^{\frac{2}{3}}}{3x^{\frac{1}{3}}(6-x)^{\frac{2}{3}}}$$

$$= \frac{-x + 2(6-x)}{3x^{\frac{1}{3}}(6-x)^{\frac{2}{3}}} = \frac{12-3x}{3x^{\frac{1}{3}}(6-x)^{\frac{2}{3}}}$$

(b) Find the local maximum and minimum values of f.

$f(4) = \sqrt[3]{4^2}\sqrt[3]{2} \approx 3.1748$ is local max. value

$f(0) = 0$ is local min. value

(c) Find the intervals of concavity and the inflection points.

$$f'(x) = \frac{1}{3}(12-3x)\left[x^{\frac{1}{3}}(6-x)^{\frac{2}{3}}\right]^{-1}$$

$$f''(x) = \frac{1}{3}(12-3x)(-1)\left[x^{\frac{1}{3}}(6-x)^{\frac{2}{3}}\right]^{-2}\left[x^{\frac{1}{3}}\left(\frac{2}{3}\right)(6-x)^{-\frac{1}{3}} + (6-x)^{\frac{2}{3}}\left(\frac{1}{3}\right)(x)^{-\frac{2}{3}}\right]$$

$$+\left[x^{\frac{1}{3}}(6-x)^{\frac{2}{3}}\right]^{-1}[-1]$$

$$f''(x) = \frac{1}{3}\left[\frac{6x^2 - 12x - (6-x)(3x-12) - 3x(6-x)}{3x^{\frac{4}{3}}(6-x)^{\frac{5}{3}}}\right] = \frac{-8}{x^{\frac{4}{3}}(6-x)^{\frac{5}{3}}}$$

Ex 4.24 (a)

When $f'(x) = 0$

$12 - 3x = 0 \iff \boxed{x = 4}$

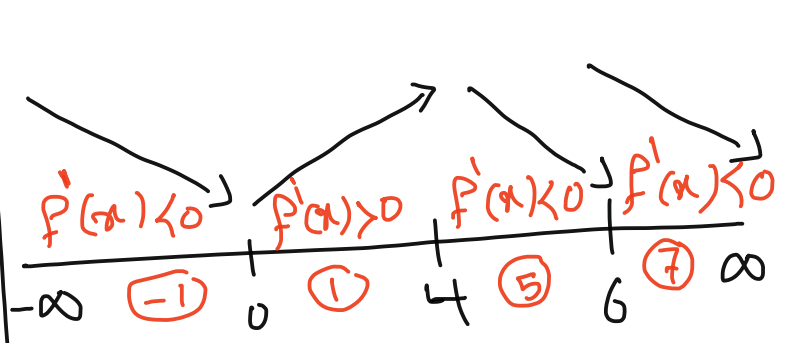
When $f'(x)$ DNE

$3\sqrt[3]{x} \quad \sqrt[3]{(6-x)^2} = 0$

Either $\sqrt[3]{x} = 0 \iff \boxed{x = 0}$

or $\sqrt[3]{(6-x)^2} = 0 \iff (6-x)^2 = 0$

$6 - x = 0 \iff \boxed{x = 6}$



$f(x)$ is increasing on $(0,4)$

$f(x)$ is decreasing on $(-\infty, 0)$, $(4, 6)$, $(6, \infty)$

Ex 4.24 (c)

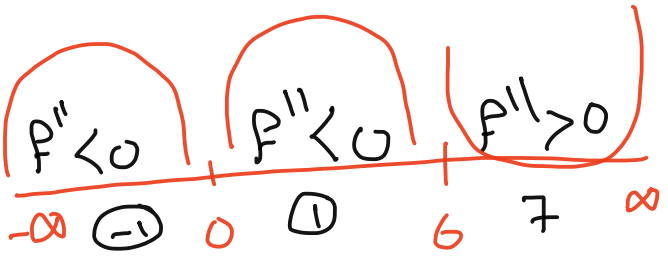
When $f''(x) = 0$ (impossible)

When $f''(x)$ DNE

$x^{\frac{4}{3}}(6-x)^{\frac{5}{3}} = 0$

Either $\boxed{x = 0}$ or

$(6-x)^{\frac{5}{3}} = 0 \iff \boxed{x = 6}$



Concave up on $(6, \infty)$

Concave down on $(-\infty, 0)$, $(0, 6)$

$(6, f(6)) = (6, 0)$ is inflection point

HW

Example 4.25. Given the function $y = e^{\frac{1}{x}}$. $D_y = \mathbb{R} \setminus \{0\}$

(a) Find the intervals on which f is increasing or decreasing.

$f(x)$ is decreasing $\forall x \in D_y$

(b) Find the local maximum and minimum values of f.

There is no local extrem values

(c) Find the intervals of concavity and the inflection points.

Concave up on $(-\frac{1}{2}, 0)$, $(0, \infty)$

Concave down on $(-\infty, -\frac{1}{2})$

$$f''(x) = \frac{e^{\frac{1}{x}}(2x+1)}{x^4}$$

Exercises Page 300 1, 9, 13, 15, 17, 21, 37(a)(b)(c), 43(a)(b)(c), 45(a)(b)(c), 47(a)(b)(c), 75, 83.

4.4 Indeterminate Forms and L'Hôpital's Rule

- **L'Hôpital's Rule**

Suppose

$$\lim_{x\to a} f(x) = 0 \quad \text{and} \quad \lim_{x\to a} g(x) = 0 \iff \lim_{x\to a} \frac{f(x)}{g(x)} = \frac{0}{0}$$

or

$$\lim_{x\to a} f(x) = \pm\infty \quad \text{and} \quad \lim_{x\to a} g(x) = \pm\infty \iff \lim_{x\to a} \frac{f(x)}{g(x)} = \mp\frac{\infty}{\infty}$$

then

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

if the limit on the right hand side exists.

Indeterminate forms: $\frac{0}{0}$, $\frac{\infty}{\infty}$, $0\cdot\infty$, $\infty-\infty$, 0^0, ∞^0, 1^∞.

Example 4.26. Use L'Hôpital's rule to evaluate $\lim_{x\to 2} \frac{x-2}{x^2-4}$ and $\lim_{x\to 0} \frac{\sin x}{x}$.

$$\lim_{x\to 2} \frac{x-2}{x^2-4} = \frac{0}{0}$$

$$\lim_{x\to 2} \frac{x-2}{x^2-4} = \lim_{x\to 2} \frac{1}{2x} = \frac{1}{4}$$

$$\lim_{x\to 0} \frac{\sin x}{x} \overset{\frac{0}{0}}{=} \lim_{x\to 0} \frac{\cos x}{1} = 1$$

Example 4.27. Evaluate $\lim_{x\to 0} \dfrac{1-\cos x}{x^2}$. $= \dfrac{0}{0}$

$$\lim_{x\to 0} \frac{1-\cos x}{x^2} = \lim_{x\to 0} \frac{\sin x}{2x} \overset{\frac{0}{0}}{=} \lim_{x\to 0} \frac{\cos x}{2} = \frac{1}{2}$$

Example 4.28. Evaluate $\lim_{x\to 1} \dfrac{\ln x}{x-1}$. $= \dfrac{0}{0}$

$$\lim_{x\to 1} \frac{\ln x}{x-1} = \lim_{x\to 1} \frac{\frac{1}{x}}{1} = \lim_{x\to 1} \frac{1}{x} = 1$$

Example 4.29. Evaluate $\lim_{x\to 0} \dfrac{\tan 2x}{\ln(1+x)}$. $= \dfrac{0}{0}$

$$\lim_{x\to 0} \frac{\tan 2x}{\ln(1+x)} = \lim_{x\to 0} \frac{2\sec^2 2x}{\frac{1}{1+x}} = \lim_{x\to 0} \frac{2(1+x)}{\cos^2 2x} = 2$$

HW

Example 4.30. Evaluate $\lim_{x\to 0} \dfrac{\sin x - x}{x^3}$. $= -\dfrac{1}{6}$

HW

Example 4.31. Evaluate $\lim_{x\to 0} \dfrac{1-\cos x}{x^2+3x}$. $= 0$

HW

Example 4.32. Evaluate $\lim_{x\to \infty} \dfrac{e^{-x}}{x^{-1}}$. $= 0$

Example 4.33. Evaluate $\lim_{x\to\infty} \frac{x^{2.5}}{e^x}$. What can you say about $\lim_{x\to\infty} \frac{x^a}{e^x}$ for any positive real number a?

Example 4.34. Show that for any positive real number a, $\lim_{x\to\infty} \frac{\ln x}{x^a} = 0$.

Example 4.35. Find $\lim_{x\to 0^+} \dfrac{\ln x}{\cot x}$.

$$\lim_{x\to 0^+} \frac{\ln x}{\cot x} = \lim_{x\to 0^+} \frac{\frac{1}{x}}{-\csc^2 x} = \lim_{x\to 0^+} \frac{-\sin^2 x}{x}$$

$$= \lim_{x\to 0^+} \frac{-2\sin x \cos x}{1} = 0$$

Example 4.36. Find $\lim_{x\to \frac{\pi}{2}} (\tan x \cdot \ln \sin x)$.

$$\lim_{x\to \frac{\pi}{2}} (\tan x \cdot \ln \sin x) = \lim_{x\to \frac{\pi}{2}} \frac{\ln \sin x}{\frac{1}{\tan x}} = \lim_{x\to \frac{\pi}{2}} \frac{\ln \sin x}{\cot x}$$

$$= \lim_{x\to \frac{\pi}{2}} \frac{\frac{\cos x}{\sin x}}{-\csc^2 x} = \lim_{x\to \frac{\pi}{2}} \frac{-\cos x \sin^2 x}{\sin x}$$

$$= \lim_{x\to \frac{\pi}{2}} -\cos x \sin x = 0$$

Example 4.37. Find $\lim_{x\to 1^+}\left(\frac{x}{x-1}-\frac{1}{\ln x}\right)$. $= \infty - \infty$

$$\lim_{x\to 1^+}\left(\frac{x}{x-1}-\frac{1}{\ln x}\right) = \lim_{x\to 1^+}\frac{x\ln x - x + 1}{(x-1)\ln x} \quad \frac{0}{0}$$

$$= \lim_{x\to 1^+}\frac{x\frac{1}{x} + \ln x(1) - 1}{(x-1)\frac{1}{x} + \ln x\,(1)} = \lim_{x\to 1^+}\frac{\ln x}{1-\frac{1}{x}+\ln x} \quad \frac{0}{0}$$

$$= \lim_{x\to 1^+}\frac{\frac{1}{x}}{+\frac{1}{x^2}+\frac{1}{x}} = \lim_{x\to 1^+}\frac{\frac{1}{x}}{\frac{1}{x}\left[\frac{1}{x}+1\right]} = \lim_{x\to 1^+}\frac{1}{\frac{1}{x}+1}$$

$$= \frac{1}{2}$$

Example 4.38. Find $\lim_{x\to\infty}(e^x - x)$. $= \infty - \infty$

$$\lim_{x\to\infty} e^x - x = \lim_{x\to\infty} e^x\left(1-\frac{x}{e^x}\right) \quad \frac{\infty}{\infty} \quad \frac{\infty}{\infty}$$

$$= \lim_{x\to\infty} e^x\left[\lim_{x\to\infty}1 - \lim_{x\to\infty}\frac{x}{e^x}\right]$$

$$= \lim_{x\to\infty} e^x\left[\lim_{x\to\infty}1 - \lim_{x\to\infty}\frac{1}{e^x}\right] = \infty[1-0] = \infty$$

$$\lim_{x\to\infty}(e^x - x) = \lim_{x\to\infty} x\left(\frac{e^x}{x}-1\right) \quad \infty \quad \frac{\infty}{\infty} \quad 1$$

$$= \lim_{x\to\infty} x\left[\lim_{x\to\infty}\frac{e^x}{x} - \lim 1\right]$$

$$= \lim_{x\to\infty} x\left[\lim_{x\to\infty}\frac{e^x}{1} - \lim 1\right]$$

$$= \infty[\infty - 1] = \infty$$

Example 4.39. Find $\lim_{x\to 0^+} (x+1)^{\cot x}$. $= (1)^{\infty}$

y

let $y = (x+1)^{\cot x}$

$\ln y = \ln (x+1)^{\cot x} = \cot x \ln (x+1)$ $\quad \infty \cdot 0$

$\lim_{x\to 0^+} \ln y = \lim_{x\to 0^+} \cot x \ln (x+1) = \lim_{x\to 0^+} \dfrac{\ln (x+1)}{\frac{1}{\cot x}} = \lim_{x\to 0^+} \dfrac{\ln (x+1)}{\tan x}$ $\quad \frac{0}{0}$

$\lim_{x\to 0^+} \ln y = \lim_{x\to 0^+} \dfrac{\frac{1}{x+1}}{\sec^2 x} = \lim_{x\to 0^+} \dfrac{\cos^2 x}{x+1} = 1$

$\lim_{x\to 0^+} \ln y = 1 \iff \lim_{x\to 0^+} y = e$

$e^{\lim_{x\to 0^+} \ln y} = e^1$

Example 4.40. Find $\lim_{x\to \frac{\pi}{2}^-} (\tan x)^{\cos x}$. $= (\infty)^{0}$

let $y = (\tan x)^{\cos x}$

$\ln y = \ln (\tan x)^{\cos x} = \cos x \ln \tan x$ $\quad 0 \cdot \infty$

$\lim_{x\to \frac{\pi}{2}^-} \ln y = \lim_{x\to \frac{\pi}{2}^-} \cos x \ln \tan x = \lim_{x\to \frac{\pi}{2}^-} \dfrac{\ln \tan x}{\frac{1}{\cos x}} = \lim_{x\to \frac{\pi}{2}^-} \dfrac{\ln \tan x}{\sec x}$ $\quad \frac{\infty}{\infty}$

$\lim_{x\to \frac{\pi}{2}^-} \ln y = \lim_{x\to \frac{\pi}{2}^-} \dfrac{\frac{\sec^2 x}{\tan x}}{\sec x \tan x} = \lim_{x\to \frac{\pi}{2}^-} \dfrac{\sec x}{\tan^2 x} = \lim_{x\to \frac{\pi}{2}^-} \dfrac{\cos^2 x}{\cos x \sin^2 x}$

$\lim_{x\to \frac{\pi}{2}^-} \ln y = \lim_{x\to \frac{\pi}{2}^-} \dfrac{\cos x}{\sin^2 x} = 0$

$e^{\lim_{x\to \frac{\pi}{2}^-} \ln y} = e^0 = 1 \iff \lim_{x\to \frac{\pi}{2}^-} y = 1$

HW

Example 4.41. Find $\lim_{x\to 0^+} x^x$. = 1

HW

Example 4.42. Find $\lim_{x\to 0^+} (\sin x)^{\cot x}$. = 0

Exercises Page 311 7, 11, 13, 15, 17, 19, 21, 23, 27, 31, 33, 35, 39, 45, 47, 49, 51, 53, 55, 57, 59, 61, 63, 79.

4.5 Summary of Curve Sketching

Step A: State the domain of the function.

- set of values of x which the function is defined.

Step B: Find the x-intercepts and the y-intercepts.

- x-intercepts: set $y = 0$.
- y-intercepts: set $x = 0$.

Step C: Check if the function is even, odd or periodic.

- even: $f(-x) = f(x)$
- odd: $f(-x) = -f(x)$
- periodic: $f(x + p) = f(x)$

Step D: Find the horizontal asymptotes and the vertical asymptotes.

- horizontal:

$$\lim_{x \to -\infty} f(x) = L \quad \text{or} \quad \lim_{x \to \infty} f(x) = L$$

then $y = L$ is the horizontal asymptote.

- vertical:

$$\lim_{x \to a^-} f(x) = \pm\infty \quad \text{or} \quad \lim_{x \to a^+} f(x) = \pm\infty$$

then $x = a$ is the vertical asymptote.

Step E: Find the intervals of increase or decrease.

- find $f'(x)$.

Step F: Find the local maximum and minimum values.

- use First Derivative Test.

Step G: Find the intervals of concavity and the inflection points.

- find $f''(x)$.

Step H: Sketch the curve.

Example 4.43. Use the guidelines to sketch the curve $f(x) = \dfrac{2x^2}{x^2 - 1}$.

Example 4.44. Sketch the graph of $f(x) = \dfrac{x^2}{\sqrt{x+1}}$.

Example 4.45. Sketch the graph of $f(x) = xe^x$.

Example 4.46. Sketch the graph of $f(x) = \dfrac{\cos x}{2 + \sin x}$.

Exercises Page 321 3, 9, 13, 39, 51.

4.7 Optimization Problems

Example 4.47. A rectangular box is made from a piece of cardboard 24 cm long and 9 cm wide by cutting out identical squares from the four corners and turning up the sides. Find the dimensions of the box of maximum volume. What is the maximum volume?

Example 4.48. A farmer has 100 meters of wire fence with which he plans to build two identical adjacent pens. What are the dimensions of the enclosure that has maximum area?

Example 4.49. Find the dimensions of the right circular cylinder of greatest volume that can be inscribed in a given right circular cone.

Example 4.50. A cylindrical can is to be made to hold 1 L of oil. Find the dimensions that will minimize the cost of the metal to manufacture the can.

Example 4.51. Find the point on the parabola $y^2 = 2x$ closest to the point $(1, 4)$.

Example 4.52. Find the area of the largest rectangle that can be inscribed in a semicircle of radius r.

Exercises Page 336 3, 5, 7, 13, 15, 23, 26, 31, 37, 73.

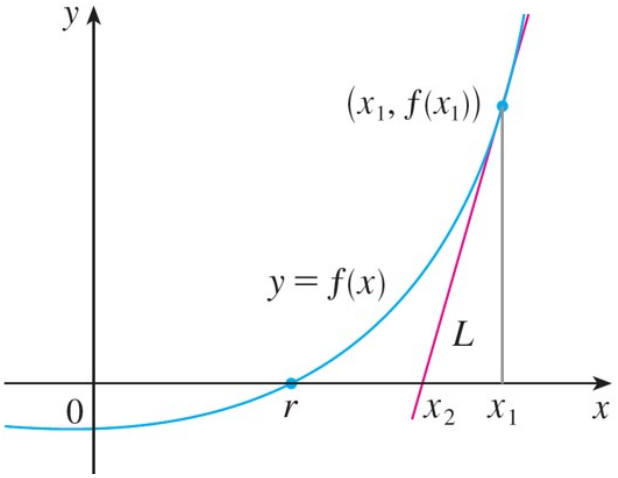
4.8 Newton's Method

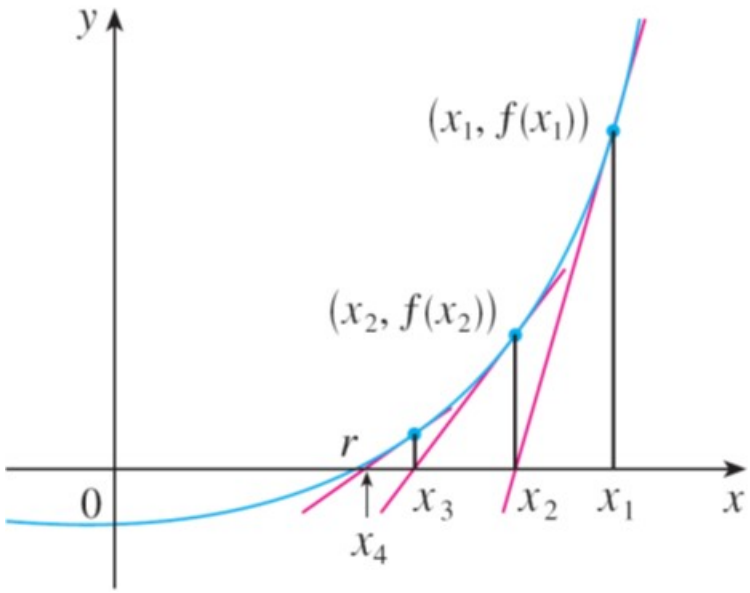
- **Newton's Method**

Let $f(x)$ be a differentiable function and let x_1 be an initial approximation to the root r of $f(x) = 0$. Let E denote a bound for the error $|r - x_n|$. Compute

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \quad n = 1, 2, \ldots$$

until $|x_{n+1} - x_n| < E$.





Example 4.53. Use Newton's Method to find the real root r of $f(x) = x^3 - 3x - 5 = 0$ to 7 decimal places.

n	x_n
1	
2	
3	
4	
5	
6	
7	
8	

Example 4.54. Use Newton's Method to find the positive real root r of $f(x) = 2 - x + \sin x = 0$ correct to 7 decimal places.

n	x_n
1	
2	
3	
4	
5	
6	
7	
8	

Example 4.55. Use Newton's method to find $\sqrt[6]{2}$ correct to 8 decimal places.

n	x_n
1	
2	
3	
4	
5	
6	
7	
8	

Exercises Page 348 15, 17, 25, 29.

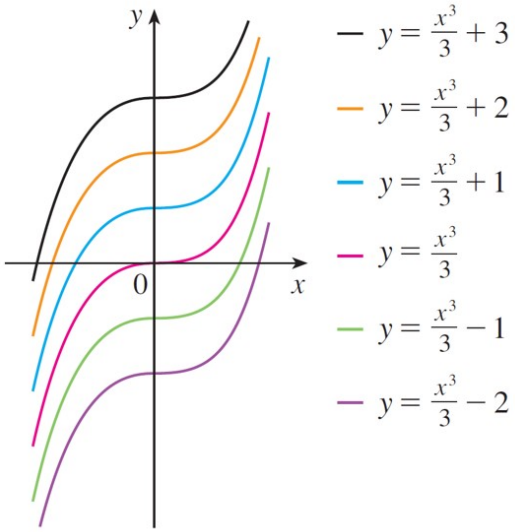
4.9 Antiderivatives

- **Definition**

We call F an **antiderivative** of f on the interval I if $D_x F(x) = f(x)$ on I, that is, if $F'(x) = f(x)$ for all x in I.

Function	Particular antiderivative	Function	Particular antiderivative
$cf(x)$	$cF(x)$	$\sin x$	$-\cos x$
$f(x) + g(x)$	$F(x) + G(x)$	$\sec^2 x$	$\tan x$
$x^n\ (n \neq -1)$	$\dfrac{x^{n+1}}{n+1}$	$\sec x \tan x$	$\sec x$
$\dfrac{1}{x}$	$\ln \lvert x \rvert$	$\dfrac{1}{\sqrt{1-x^2}}$	$\sin^{-1} x$
e^x	e^x	$\dfrac{1}{1+x^2}$	$\tan^{-1} x$
b^x	$\dfrac{b^x}{\ln b}$	$\cosh x$	$\sinh x$
$\cos x$	$\sin x$	$\sinh x$	$\cosh x$

Example 4.56. Find an antiderivative of the function $f(x) = x^2$ on $(-\infty, \infty)$.



If $F(x)$ is an antiderivative of $f(x)$, so is $F(x) + C$ for any constant C.

Example 4.57. Find the antiderivative of $f(x) = 4x^3$ on $(-\infty, \infty)$.

Example 4.58. Find all the functions g so that

$$g'(x) = 4\sin x + \frac{2x^5 - \sqrt{x}}{x}$$

Example 4.59. Find f if $f'(x) = e^x + \dfrac{20}{1+x^2}$ and $f(0) = -2$.

Example 4.60. Find f if $f''(x) = 12x^2 + 6x - 4$ and $f(0) = 4$, $f(1) = 1$.

Example 4.61. A particle moves in a straight line and has acceleration given by $a(t) = 6t + 4$. Its initial velocity is $v(0) = -6$ cm/s and its initial displacement is $s(0) = 9$ cm. Find its position function $s(t)$.

Exercises Page 355 7, 11, 17, 21, 33, 47, 63.